



# A novel phase-based video motion magnification method for non-contact measurement of micro-amplitude vibration

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## ABSTRACT

Vibration analysis is crucial for structural health monitoring and fault diagnosis. Conventional contact sensors present limitations, prompting the adoption of non-contact methods such as laser Doppler vibration measurements and computer vision-based techniques. Among these, phase-based video motion magnification has gained prominence for its high resolution and ability to capture comprehensive vibration data across the entire field. However, traditional video motion magnification methods are often affected by noise and artifacts when facing micro vibration measurements. Especially at high magnification, artifacts such as "double edges" often appear, which seriously affects the accuracy of vibration analysis. In addition, for complex structures, the existing methods still have some difficulties in extracting vibration modes and maintaining fine motion details. Therefore, we propose a novel method for micro-amplitude vibration magnification and a combination of the Horn–Schunck method and the motion intensity averaging method for non-contact vibration displacement measurement. The proposed method uses a complex steerable pyramid to decompose the video into multi-scale and multi-directional sub-bands, extract the phase variations of interest and magnify them, and then an optimized one-dimensional row-gradient domain-guided image filter is used to finely eliminate the double edges, artifacts, and other noises in the high-frequency sub-bands, and finally synthesize the motion-magnified video. Experimental validations on a vibration platform and real structural elements, including a three-story building and an aluminum cantilever beam, demonstrate the method's superiority in preserving structural clarity and minimizing information loss. Our approach significantly enhances structural vibration analysis, offering accurate frequency identification and mode shape extraction.

## 1. Introduction

Vibration signal analysis is one of the primary techniques used in structural health monitoring (SHM) [1] and the diagnosis of mechanical system faults [2]. In engineering practice, many mechanical faults are caused by micro-amplitude vibrations [3,4], it is essential to accurately measure the micro-amplitude vibration of mechanical structures [5–8].

In recent decades, various sensors and data processing techniques have been proposed and applied to measure the vibration displacement of structures. Traditional vibration measurements use contact sensors, such as accelerometers, but contact sensors need to be mounted on the surface of the structure, which can affect the structure's dynamic properties due to increased loading [9]. Moreover, contact sensors can only provide sparse measurements at a limited number of discrete locations, restricting their ability to capture comprehensive vibration data across the entire structure. To address these issues, non-contact vibration measurement

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techniques have been developed, such as laser Doppler vibrometry [10], eddy current sensors [11], microwave interferometry [12], and vision-based methods [13,14].

With the advancements in visual instruments, including high-resolution cameras and camcorders, alongside the progress in image and video processing techniques, vision-based measurements have become increasingly prevalent in mechanical engineering. Furthermore, compared to traditional methods, vision-based measurement techniques offer several advantages, including relatively low cost, flexibility, high resolution, absence of additional load requirements, and the ability to capture full-field vibration data [15].

There are different algorithms that have been developed to apply to this technology. The most frequently employed algorithms include the template matching algorithm [16] and digital image correlation (DIC) [17]. Template matching methods allow tracking the displacement of the region of interest (ROI) along the video frame sequence [18], DIC algorithms are usually more accurate than template matching [17], but they require the setting of a scattering pattern on the surface to be analyzed, edge detection techniques are also widely used and are employed in areas such as bridge vibration analysis [19] and pipe vibration analysis [20]. These methods are suitable for objects with clear contours and can capture the edge motion of the object, but they cannot provide accurate motion information for objects with unclear texture or blurred edges. In real-world scenarios, the vibration amplitude of mechanical structures is usually in the minor range, which limits the practical applicability of the target matching tracking algorithm.

Vibration measurements based on the optical flow method estimate vibration displacements by comparing the movement of pixels in consecutive frames of an image. It provides pixel-level accurate motion field information that captures the motion of each pixel point in the image and can be used to capture complex motion [21–23]. It is mainly categorized into sparse optical flow method and dense optical flow method. Among them, the sparse optical flow method tracks the motion of selected specific pixel points, which is used on the premise that the luminance between neighboring frames is constant, the motion of the tracked pixel points between neighboring frames is relatively small, and the pixels in the selected specific region have consistent motion [17]. In contrast, the dense optical flow method estimates motion signals for each pixel point in the image to obtain the motion field of the whole image, which does not need to satisfy the premise of spatial consistency of the pixels and can be applied to complex structures with irregular geometries or coupling patterns [12,24,25].

Recently, the phase-based video motion magnification (PVMM) method has emerged and has been employed to unveil small or even imperceptible periodic phenomena in the real world. The PVMM algorithm achieves image decomposition by utilizing a complex steerable pyramid [26]. PVMM has been applied to capture micro-amplitude vibrations of structures [27], detect vibrations in cantilever girders [28–31], buildings [13,32,33], light poles [34], antenna towers [35], wind turbine blades [36], arch dams [37], in-service steel bridges [35,38], and cables [39] by analyzing modal frequencies and modal vibration shapes. Laboratory experiments have validated the PVMM algorithm's ability to effectively magnify small structural motions within the frequency region of ROI [13,28–30]. In combination with DIC, it can successfully extract the operational deflected shapes of cantilever beams from the magnified videos. When noise is present in the video, combining PVMM with DIC can substantially enhance the accuracy of identifying the structural modal vibration shapes [40]. A broadband phase video motion magnification and line tracking algorithm based on this method was proposed and applied to cable vibration measurements with micro amplitudes, which utilizes a wide bandwidth for processing and improves computational efficiency [41]. The high modal order motion assessment capability of the method can be enhanced by optimizing the phase correction [42], and the micro-amplitude vibrations captured by the method can be used for different structural health monitoring and damage level assessment [31,33]. The wide range of applications in the mechanical field illustrates that PVMM can significantly magnify the motion of vibrations, which does not require additional training time and does not produce noticeable distortions at large magnifications, but the conventional PVMM above produces artifacts that affect the accuracy of subsequent vibration mode and frequency measurements. Recently, the deep learning-based video magnification method (LB-VMM) has been designed to magnify subtle signals within videos [43]. It utilizes Convolutional Neural Networks (CNNs) for processing and has been demonstrated to magnify vibrations within videos, enabling damage detection on laboratory structural benchmarks [44], the Swin Transformer was applied to LB-VMM process to reduce the noise generation, and it can also be used to measure micro-amplitude vibrations in different scenarios [45]. However, these learning-based methods currently necessitate entirely synthetic datasets and require additional substantial training time. Moreover, when utilizing higher magnification factors, videos tend to exhibit different degrees of distortion.

This paper presents a new video motion magnification method and vibration measurement technique capable of accurately capturing micro-amplitude vibration signals. The video magnification method utilizes a complex steerable pyramid and an optimized one-dimensional row-gradient domain-guided image filter (1D Row GDGIF) to effectively magnify micro-amplitude vibrations while maintaining image quality. The vibration measurement method employs the Horn–Schunck method and motion intensity averaging techniques to provide highly accurate vibration signal measurements. The aim of this paper is to provide new and efficient solutions in the field of micro-amplitude vibration measurement.

We have established a vibration testing platform to perform experiments on micro-amplitude vibration measurements. Then we performed several vibration magnification and measurement experiments on real structural elements: a three-story building structure and an aluminum cantilever beam structure. The remainder of this paper is structured as follows: Section 2 elaborates on the theory behind the constructed framework's strategy, Section 3 explores the identification of vibration displacements and vibration frequencies of structures with different shapes using our proposed framework on a vibration platform. Experiments on vibration measurements on real structural elements, a three-story building structure and an aluminum cantilever beam structure, are then presented. Discussion and conclusions are presented in Section 4, along with an outlook on the future development of the current framework.

## 2. The proposed method

In this section, we present the theoretical framework of our proposed method for micro-amplitude vibration identification. We have chosen to convert the image from RGB color space to YIQ (NTSC) color space. This is because the YIQ space better separates the luminance information (Y-channel) and the chrominance information (I and Q-channels) of an image. Choosing this color space helps to better capture the structural information of the image and reduces the impact of color information on the image processing, thus helping the PVMM and optimized 1D Row GDGIF processes to produce accurate and reliable results.

### 2.1. Phase-based video motion magnification

Motivated by a phase-based approach for estimating optimal flow fields, we employ the complex steerable pyramid in video processing. This allows us to apply temporal and spatial filters to select the appropriate frequency signals, compute motion information, and perform motion magnification [26]. The complex steerable pyramid algorithm consists of quadrature phase filters similar to those provided by complex Gabor filters. These non-aliased sub-band filters can decompose an image into a range of spatial scales and orientations.

We apply the complex steerable pyramids to each video frame  $f_t(x, y)$  at spatial location  $(x, y)$  and time  $t$  to obtain the sub-band representation  $s$  in scale  $r$  and orientation  $\theta$  as follows:

$$s_{r,\theta,t}(x, y) = \psi_{r,\theta} \otimes f_t(x, y) = a_{r,\theta,t}(x, y)e^{i\phi_{r,\theta,t}(x,y)}, \quad (1)$$

where  $a$  is local amplitude,  $\phi$  is a local phase,  $\psi_{r,\theta}$  is the basic filter, and the symbol  $\otimes$  stands for convolution. The displaced frame is composed of the total sub-bands and defined as:

$$f_t((x, y) + \delta_t(x, y)) = \sum_{r,\theta} s_{r,\theta,t}((x, y) + \delta_t(x, y)), \quad (2)$$

where  $\delta_t(x, y)$  is a displacement function.

The phase difference  $\phi\delta_t(x, y)$  of  $\delta_t(x, y)$  can be obtained by the following formula:

$$\phi\delta_t(x, y) = \arg(a_{r,\theta,t}e^{i\phi((x,y)+\delta_t(x,y))}) - \arg(a_{r,\theta,t}e^{i\phi(x,y)}). \quad (3)$$

We can get the total phase difference representation  $\Delta\phi(x, y)$  through  $\phi\delta_t(x, y)$ :

$$\Delta\phi(x, y) = \sum_{r,\theta} \phi\delta_t(x, y). \quad (4)$$

The phase difference  $\Delta\phi$  is then multiplied by a magnification factor  $\alpha$  to increase the phase of sub-bands  $s_{r,\theta}$  by this amount to get the motion magnified sub-bands as:

$$\hat{s}_{r,\theta,t}(x, y) = s_{r,\theta,t}(x, y)e^{i\alpha\Delta\phi} = a_{r,\theta,t}(x, y)e^{i(\phi+\alpha\Delta\phi)}. \quad (5)$$

### 2.2. Optimized one-dimensional row-gradient domain-guided image filter

The quality of texture and motion information in video magnification algorithms is crucial, and we have found that stripe noise and double edge artifacts are the main sources of noise in practical applications, so we have introduced and optimized the idea of an optimized 1D Row GDGIF method, and unlike conventional 2D guided filtering using a square window, our approach utilizes a localized row window, which is more suitable to isolate the stripe component in high-frequency information images.

Our proposed method is the first video motion magnification method that involves an organic combination of complex domain processing and gradient domain processing, and achieves an effective use of their respective advantages by the proposed algorithm in this paper. Through the decomposition of complex manipulable pyramids, we obtain sub-bands at different scales and orientations, and then generate new sub-bands by using phase-difference magnification, this complex-domain processing method has good sensitivity to small vibrations. Next, we combine the sub-bands in different directions at the same scale to obtain high-frequency and low-frequency sub-band images after motion magnification. On the high-frequency sub-band images, we introduce an optimized 1D Row GDGIF method for the filtering process, which combines the processing advantages of the gradient-domain and reduces the artifacts that may occur during motion magnification, while maintaining the clarity and structural contour features of the video.

Next equation represents the local linear model of guided filtering, establishing a local linear correlation between the filtered output within the local row window and the guide image:

$$H_i = \bar{a}_{rk}u_i + \bar{b}_{rk}, \forall i \in \omega_{rk}, \quad (6)$$

where  $u_i$  is the phase magnified high-frequency sub-band image, which is also the guide image,  $H_i$  is the output image filtered using the guide image,  $\omega_{rk}$  is the square window with the central pixel  $k$ ,  $i$  is used as an index to iterate through each pixel within the window,  $\bar{a}_{rk}$  and  $\bar{b}_{rk}$  are the average values of the sum of linear coefficients respectively, and its solution is provided by the following formula:

$$\begin{aligned} \bar{a}_{rk} &= \frac{1}{|\omega_{rk}|} \sum_{k \in \omega_{rk}} a_{rk} \\ \bar{b}_{rk} &= \frac{1}{|\omega_{rk}|} \sum_{k \in \omega_{rk}} b_{rk} \end{aligned}, \quad (7)$$

where  $|\omega_{rk}|$  is the number of pixels in the local row window  $\omega_{rk}$ . The optimal values of the linear coefficients  $a_{rk}$  and  $b_{rk}$  are obtained by minimizing the cost function  $E$ , which is defined as:

$$E = \sum_{i \in \omega_{rk}} \left[ (a_{rk}u_i + b_{rk} - u_i)^2 + \frac{\mu}{\Gamma_{u_i}} (a_{rk} - \gamma_{rk})^2 \right], \quad (8)$$

where  $\mu$  is a scalar and its value will affect the output result. The definition of  $\gamma_{rk}$  is as follows:

$$\gamma_{rk} = 1 - \frac{1}{1 + e^{\eta(\chi(rk) - \xi_{\chi, \infty})}}, \quad (9)$$

where,  $\chi(rk)$  is defined as  $\sigma_{u,1}(rk)\sigma_{u,16}(rk)$ , the size of the filter window is set as 16, and  $\sigma_{u,1}(rk)$  is the standard deviation of the image  $u$  in the local row window  $\omega_1(rk)$ , the value of  $\eta$  is set as  $4/(\xi_{\chi, \infty} - \min(\chi(i)))$ ,  $\xi_{\chi, \infty}$  is the average of  $\chi(i)$ . The value of  $\gamma_{rk}$  close to 1 indicates that the pixel  $k$  is in the edge area, if the pixel is in the smooth area, the value of  $\gamma_{rk}$  will be close to 0. Therefore, the edge preserving ability of gradient domain guided filtering is better than the traditional guided filtering algorithm.

In addition, the parameter  $\Gamma_{u_i}$  is edge perception weight, which is defined as:

$$\Gamma_{u_i} = \frac{1}{N} \sum_{i=1}^N \frac{\chi(c_k) + \varepsilon}{\chi(i) + \varepsilon}, \quad (10)$$

where  $N$  is the number of pixels in the image  $u_i$ ,  $\varepsilon$  is a very small value to prevent the denominator from being 0. The edge perception weight can judge the importance of each pixel to the guide image, and thus reduce the misjudgment of non-noise pixels as noise, and further improve the image detail retention ability of the algorithm.

Accordingly, the solution of the optimal value of linear coefficient  $a_{rk}$  and  $b_{rk}$  is:

$$a_{rk} = \frac{\xi_{u,*X,16}(rk) - \xi_{u,16}(rk)\xi_{X,16}(rk) + \frac{\mu}{\Gamma_{u,16}(rk)}\gamma_{ck}}{\sigma_{u,16}^2(rk) + \frac{\mu}{\Gamma_{u,16}(rk)}}, \quad (11)$$

$$b_{rk} = \xi_{X,16}(rk) - a_{rk}\xi_{u,16}(rk)$$

where  $*$  represents the matrix point multiplication,  $\sigma_{u,16}^2(rk)$  is the variance of the guide image  $u$  in the local row window  $\omega_{rk}$ ,  $\xi_{u,*X,16}(rk)$ ,  $\xi_{u,16}(rk)$  and  $\xi_{X,16}(rk)$  are the mean values of  $u \cdot X$ ,  $u$  and  $X$  in the local row window  $\omega_{rk}$ , respectively.

The optimized 1D Row GDGIF not only preserves image details and edges through gradient-domain guided filtering, but also reduces computational complexity and enhances the efficiency of separating streak noise from high-frequency information images. This approach improves the quality and visualization of the magnified video while maintaining image stability and clarity, making the magnified video suitable for accurate analysis and measurement.

### 2.3. Vibration signal acquisition

The vibration measurement method proposed in this paper integrates the Horn-Schunck method and motion intensity averaging techniques to precisely capture micro-amplitude vibration signals in structural elements. Firstly, standard video frames are reconstructed using filtered high-frequency sub-band images and low-frequency sub-bands, and then, the video frames are converted to gray-scale images using a dense optical flow method, and pixel-level motion variations are captured by identifying the motion information of all the pixel points within the ROI. Finally, the pixel motion intensities within the ROI are integrated by averaging the motion information of these pixel points to obtain accurate and reliable vibration information.

Initially, after identifying a ROI  $R$  within the video, consecutive frames undergo processing using the Horn-Schunck method [12]. This process acquires motion information from all pixel points within  $R$ , capturing pixel-level motion variations. Subsequently, the intensities of motion information  $I$  from these pixel points are averaged within  $R$ , consolidating the pixel motion intensities and resulting in the derivation of motion intensity values  $I_R$ :

$$I_R = \frac{1}{N_R} \sum_{p=1}^{N_R} I_p, \quad (12)$$

where  $N_R$  is the number of pixel points in  $R$ ,  $I_p$  is The motion intensity of pixel  $p$ .

Setting the number of video frames to be  $f$ , we make use of the average motion intensity  $I_R$  to compute the resulting displacement profile equation  $D(t)$ , which represents the displacement of  $R$  at time  $t$ :

$$D(t) = I_R \times t \times \frac{1}{f}. \quad (13)$$

FFT is a method of converting a time domain signal into a frequency domain signal. By applying the FFT transformation to the displacement curve equation  $D(t)$ , we can convert the displacement over time curve into an amplitude versus frequency representation  $Y(f)$ :

$$Y(f) = FFT(D(t)), \quad (14)$$

where  $Y$  denotes the amplitude information at different frequencies  $f$ .

The major frequency components are presented as peaks in the spectrogram, indicating the most dominant vibration frequencies in the motion of the object.

The overall flowchart of the proposed method in this paper is shown in Fig. 1.

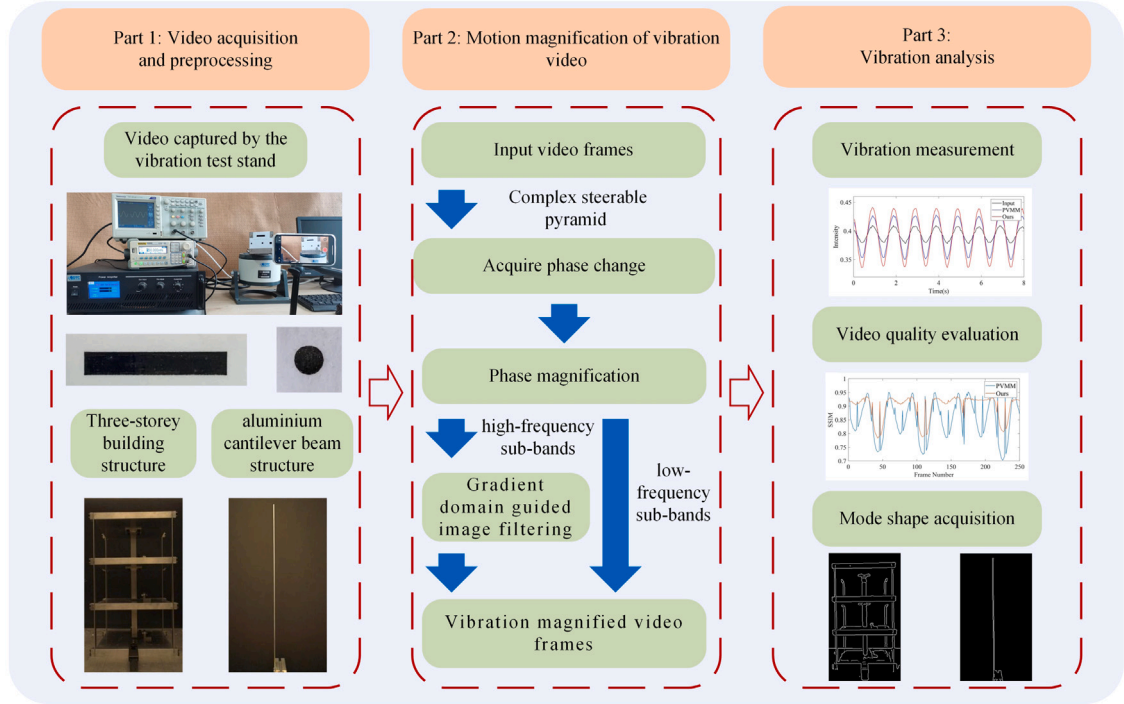


Fig. 1. Overall flow chart of the proposed method.

### 3. Experimental validation

In this section, experiments on magnification and measurement of micro-amplitude vibrations of metal objects with different shape markings having standard vibration waveforms, as well as experiments on vibration measurements and modal analysis of real structural elements: a three-story building structure and an aluminum cantilever beam structure, are performed. We discuss the selection of parameters, the use of non-contact vibration video measurement algorithms. Then evaluation metrics including vibration time and frequency domain analysis, edge detection analysis, objective evaluation metrics for video quality, and modal vibration pattern extraction analysis of complex structures are proposed, together with validation of the effectiveness of the proposed method.

#### 3.1. Shaker-based vibration visual measurement experiment

In this section, we detailed the shaker-based vibration vision measurement experiment, which serves as a key fundamental validation of our proposed methodology. Here, we focus on the experimental setup specifically designed to validate our approach. This includes an in-depth discussion of parameter selection, extraction of vibration displacements and frequencies, and the results derived from our non-contact vibration video measurement algorithm.

##### 3.1.1. Experimental setup

To assess the performance of the proposed method in this paper, we established an experimental platform based on a shaker. The Fig. 2 illustrates the composition of the entire experimental platform, which includes a shaker (TEN-V150) manufactured by Dynatronic Corporation, a signal generator (DG1022U), a power amplifier (DTCLA-200), and a digital storage oscilloscope (TDS 2012C). The shaker's moving elements are supported by a robust linear guidance system consisting of uniquely constructed radial and transverse flexure bars. These flexure rods are constructed from a bonded sandwich of spring steel and damped elastomers to provide crisp acceleration waveforms with minimal cross-motion and low distortion characteristics. A contact accelerometer sensor (model YD-31D) was arranged on the vibration table and a constant current power supply (model GD-204) was used to power the accelerometer. A portable iPhone 11 was employed as a camera to capture the vibration video. We implemented these methods in MATLAB version 2020a, and the experimental results were generated using unoptimized MATLAB code on a desktop computer equipped with an AMD Ryzen 5 4600H CPU and 16 GB RAM.

Initially, we affixed a metal block to the shaker and generated consistent sinusoidal vibrations using a signal generator and a power amplifier. Rectangular and circular markings were applied to the metal block to assess the versatility of our method. The structure was recorded using an iPhone 11 camera operating at a frame rate of 30 frames per second with a resolution of  $1080 \times 1920$  pixels. The linear distance between the shaker and the camera was approximately 250 mm. The illumination conditions were typical indoor lighting, with no external lighting enhancements.

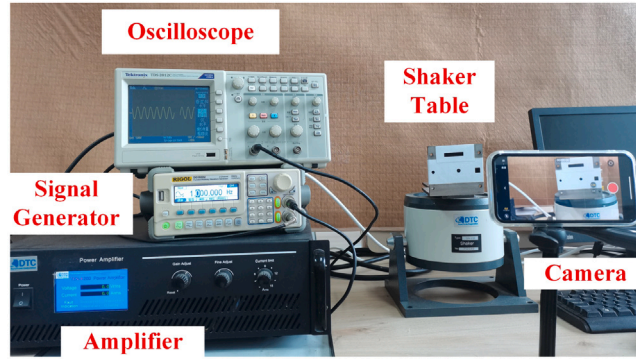


Fig. 2. Shaker-based experimental platform for visual measurement of vibration.

Table 1  
Parameters for four videos.

| Video   | Marker shape | Size ( $H \times M$ ) | Number of frames | Vibration frequency of shaking table (Hz) |
|---------|--------------|-----------------------|------------------|-------------------------------------------|
| Video 1 | Rectangle    | 200*640               | 250              | 0.5                                       |
| Video 2 | Rectangle    | 200*640               | 250              | 1                                         |
| Video 3 | Round        | 150*150               | 250              | 0.5                                       |
| Video 4 | Round        | 150*150               | 250              | 1                                         |

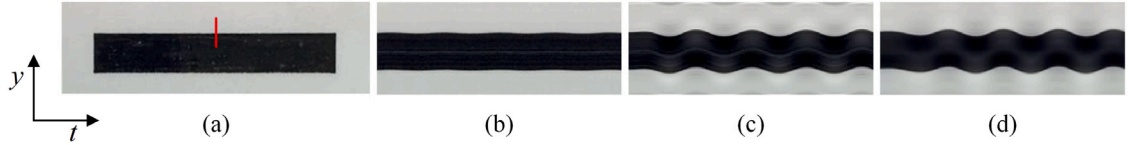


Fig. 3. Space-time slice diagram of video 1. (a) Video slice position, (b) Input, (c) PVMM, (d) Ours.

The captured raw video data is extensive, consuming a considerable amount of memory space and demanding significant computational time for processing. Given that the predominant portion of the field of view in each video frame contains background, we implemented preprocessing steps, such as cropping and intercepting the raw video, to preserve the vibration information containing the markers. Consequently, based on the diverse shapes and vibration frequencies of the markers, we generated four video sequences for subsequent experiments in video vibration measurement, as outlined in Table 1.

The pre-processed video is magnified by our proposed new video magnification method, the magnification factor is set to 10, the interest bands of video 1 and video 3 are preset to [0.3 Hz, 0.7 Hz], and the interest bands of video 2 and video 4 are preset to [0.8 Hz, 1.2 Hz]. In the process of the optimized 1D Row GDGIF, we set the size of the localized row window to a constant  $h = 16$ , and the parameter  $\mu$  is set to 0.022.

### 3.1.2. Vibration measurement results

To visualize the effect of video motion magnification processing, we have generated spatio-temporal slice diagrams, as shown in Figs. 3 to 6. Fig. 3(a) is the first frame of video 1 before processing. The red line represents the position of the selected slice, where  $t$  is the time axis, and  $y$  is the slice of each picture frame at the red line. Fig. 3(b) depicts the change trajectory at the original video slice, which is also the motion trajectory. Fig. 3(c) displays the change trajectory after the PVMM, and Fig. 3(d) illustrates the motion trajectory after our proposed method. Figs. 4 to 6 showcase the slice positions and spatio-temporal slice comparison graphs for video 2, video 3, and video 4, respectively. Figs. 4 to 6(c) presents the spatio-temporal slice plots after the PVMM, while Figs. 4 to 6(d) exhibits the spatio-temporal slice plots after our method. From Figs. 3 to 6(b), it is evident that the vibrations of the rectangular markers in the original videos 1 and 3, as well as the circular markers in the original videos 2 and 4, are faint and challenging to detect with the naked eye. After processing by PVMM and our methods, the vibrations become readily observable with increased amplitude. Most importantly, our method substantially reduces artifacts within the markers, at the marker-background junction, and in the background itself. This results in vibrations with the highest amplitude and enhanced visibility.

In Fig. 7, we have selected the ROI in the vibration videos containing rectangular and circular markers, respectively. We extracted the motion intensity value of the regions as the vibration displacement signals. Fig. 8 illustrates that both the traditional video magnification method and our approach magnify the vibration amplitude, and the resulting vibration waveforms align with the sinusoidal waveforms of the corresponding frequencies generated by the shaker. Notably, in comparison to the traditional method, our approach yields a smoother and clearer displacement curve with a larger amplitude. This indicates that our method not

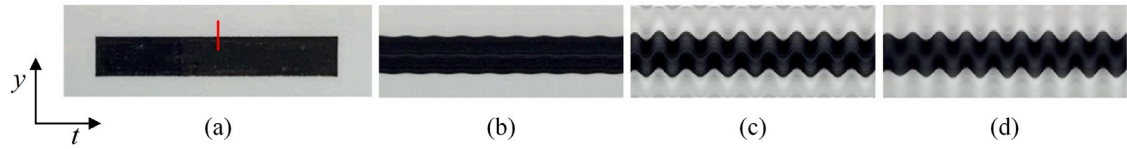


Fig. 4. Space-time slice diagram of video 2. (a) Video slice position, (b) Input, (c) PVMM, (d) Ours.

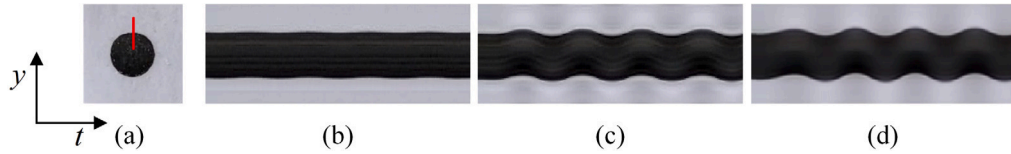


Fig. 5. Space-time slice diagram of video 3. (a) Video slice position, (b) Input, (c) PVMM, (d) Ours.

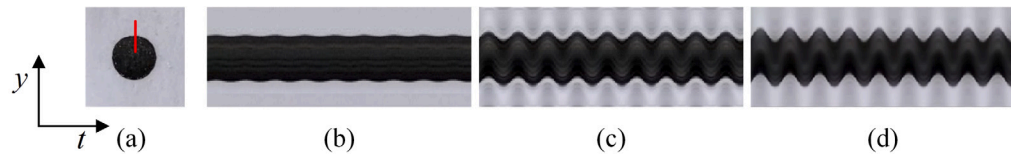


Fig. 6. Space-time slice diagram of video 4. (a) Video slice position, (b) Input, (c) PVMM, (d) Ours.



Fig. 7. Region of interest for displacement extraction.

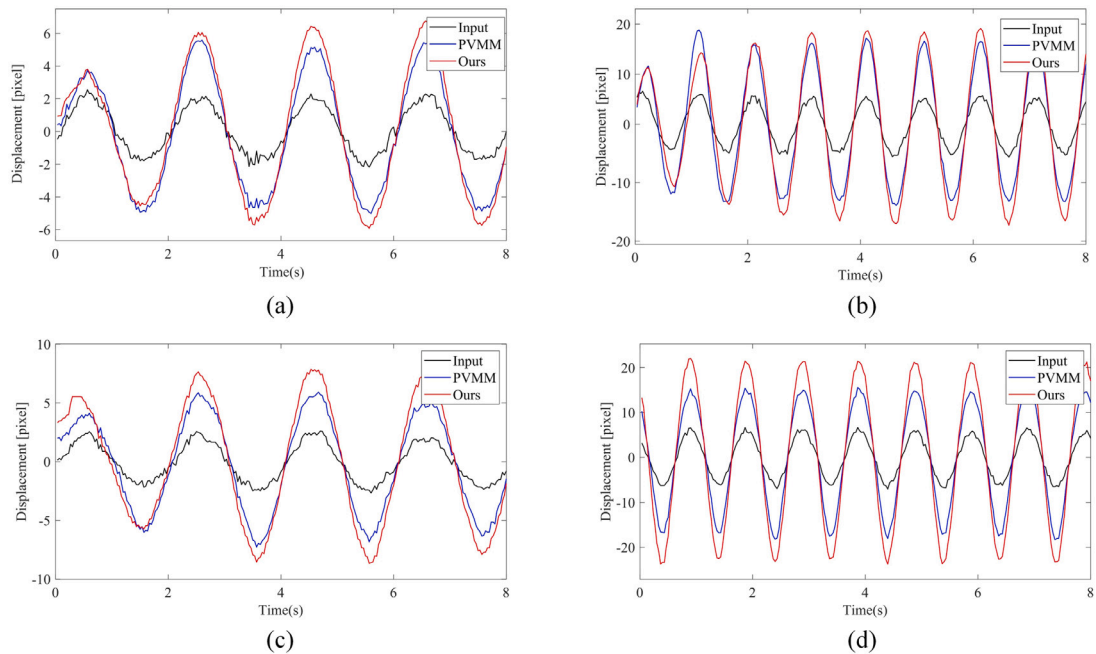


Fig. 8. Vibration displacement extraction from video. (a) Video 1, (b) Video 2, (c) Video 3, (d) Video 4.

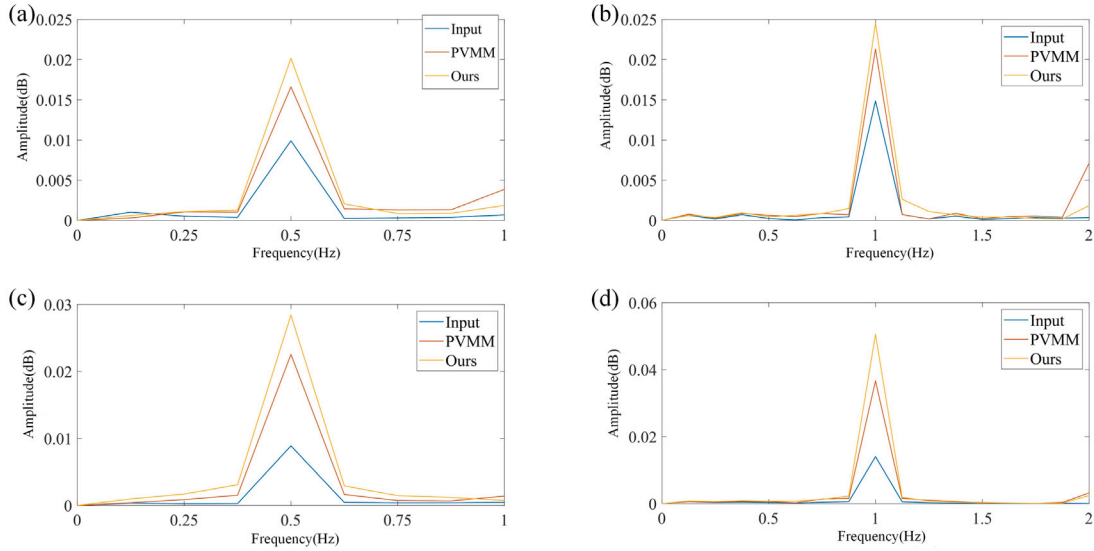


Fig. 9. Vibration frequency identification in videos. (a) Video 1, (b) Video 2, (c) Video 3, (d) Video 4.

Table 2

Frequency peak amplitude (dB) in videos.

| Video   | Input    | PVMM    | Ours           |
|---------|----------|---------|----------------|
| Video 1 | 0.03792  | 0.07473 | <b>0.08566</b> |
| Video 2 | 0.01487  | 0.02131 | <b>0.02457</b> |
| Video 3 | 0.008907 | 0.02255 | <b>0.02841</b> |
| Video 4 | 0.01407  | 0.03668 | <b>0.05052</b> |

only mitigates the artifactual noise interference associated with video motion magnification but also achieves excellent results in magnifying and identifying subtle vibrations.

We conducted an FFT analysis of the vibration frequency of the metal block, as illustrated in Fig. 9. The peak amplitude obtained for each video processing method corresponds to the frequency of the input sine wave from the shaker. Meanwhile, the proposed non-contact measurement method agrees with the vibration frequency captured by the accelerometer. In videos 1 and 3, the vibration frequency measured by the contact accelerometer is 0.5 Hz, and in videos 2 and 4, the vibration frequency measured by the contact accelerometer is 1 Hz, which is consistent with the vibration frequency measured by the proposed non-contact measurement method for the videos. This indicates the accuracy of the video magnification methods in identifying vibration frequencies (0.5 Hz and 1 Hz). Specific peak amplitudes for vibration frequencies are presented in Table 2. The peak amplitudes of the original video input vibrations are minimal. While the conventional PVMM method enhances the peak amplitudes of vibration frequencies, our method achieves the highest peak amplitudes among all four videos. This verifies the efficiency of our method in separating vibration signals, which is valuable for modal identification and SHM in mechanical structures.

### 3.1.3. Vibration video quality objective evaluation results

The quality of video after the video magnification process plays a crucial role in the accuracy of measurements for various applications relying on magnified video sequences. For quantitative comparison of motion magnified video quality, we selected the video quality metrics structural similarity (SSIM) and peak signal-to-noise ratio (PSNR) [46,47]. SSIM is derived by assessing the similarity of three components: luminance, contrast, and structure, concerning the original image. PSNR, on the other hand, calculates the maximum intensity value of Y-channel pixels (255) using the YIQ space of the base image. It is computed by dividing 10 by the logarithm of the mean square error value and then multiplying by 10. The SSIM and PSNR values for each frame of the video are calculated and averaged to yield objective evaluation index parameter values for the video.

An SSIM value of 1 implies identical similarity to the original image, while a value of 0 signifies no similarity at all. Larger SSIM values indicate better image quality. Similarly, higher PSNR values indicate excellent video quality, with reduced distortion and noise. Fig. 10 displays the SSIM values for each frame in videos 1 to 4. As observed in Fig. 10, the SSIM values of processed video frames consistently increase, and fluctuations decrease significantly in our method compared to the traditional method. This suggests that our method results in higher clarity and improved visibility in the processed video. Table 3 presents the average SSIM and PSNR values for all frames of motion-magnified videos processed using various methods. Our method outperformed other methods in terms of evaluation metrics for each video sequence. The overall highest evaluation metrics were achieved in video 3, with an SSIM value of 0.9222 and a PSNR value of 35.2236. This objectively demonstrates the effectiveness of our method in video motion magnification by reducing noise and artifact generation, yielding excellent results.

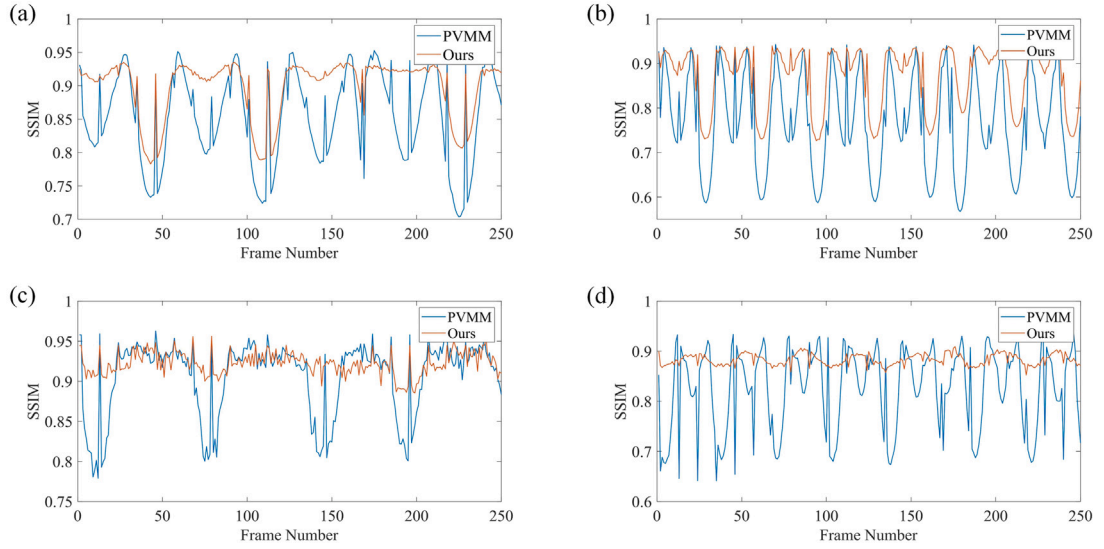


Fig. 10. SSIM comparison result chart of each frame in the video. (a) Video 1, (b) Video 2, (c) Video 3, (d) Video 4.

**Table 3**  
Comparison of objective evaluation indicators of video quality.

| Video   | PVMM (SSIM, PSNR) | Ours (SSIM, PSNR)      |
|---------|-------------------|------------------------|
| Video 1 | 0.8549, 31.6692   | <b>0.899, 32.1785</b>  |
| Video 2 | 0.7688, 27.6265   | <b>0.8699, 29.0493</b> |
| Video 3 | 0.9052, 33.0194   | <b>0.9222, 35.2236</b> |
| Video 4 | 0.8276, 28.371    | <b>0.8613, 28.5446</b> |

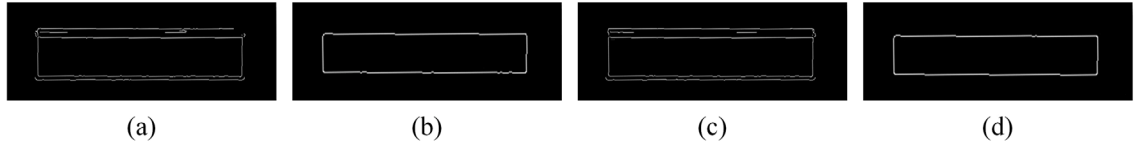


Fig. 11. Comparison graph of edge detection results of rectangular markers. (a) Keyframe 1 after PVMM, (b) Keyframe 1 after Ours, (c) Keyframe 2 after PVMM, (d) Keyframe 2 after Ours.

### 3.1.4. Edge detection results

In our study, we further investigated the situation of the edge portion of the motion magnification video, like existence of double edge phenomenon, and performed edge detection experiments. To ensure the accuracy and comparability of the results, we chose the Canny edge detection algorithm, which is effective in detecting image edges. We selected representative frames from the motion video for processing. We compare the PVMM method with the method in this article, perform Canny edge detection on them respectively, obtain the edge information of the image, and compare the edge detection results of PVMM and this method. We observed and recorded the presence or absence of the double edge phenomenon and the differences in edge detection between the different methods. As shown in Figs. 11 and 12, in our experiments, we found that the traditional phase method showed a clear double edge phenomenon, while our proposed method did not show this phenomenon. Specifically, the detection of falsified double edges using the conventional phase method demonstrates the method's introduction of redundant edge information at the edges. In contrast, our method provides better suppression of this phenomenon and reduces the generation of edge artifacts. This result clearly demonstrates the superiority of our method in avoiding the phenomenon of double edges and confirms the ability of our method to reduce the associated artifacts with double edge. The rectangular and circular markers in our setup provide further evidence that our method works well for different edge shapes in suppressing the double edge effect.

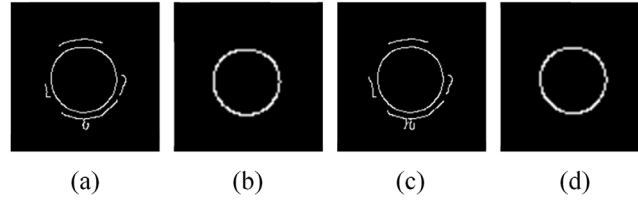


Fig. 12. Comparison graph of edge detection results of circular markers. (a) Keyframe 3 after PVMM, (b) Keyframe 3 after Ours, (c) Keyframe 4 after PVMM, (d) Keyframe 4 after Ours.

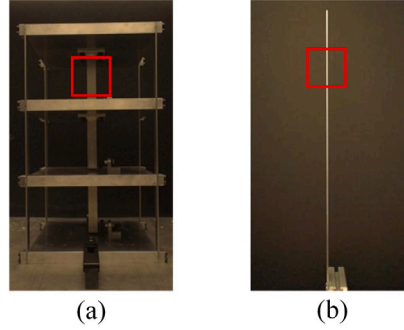


Fig. 13. Region of interest for displacement extraction of structural elements. (a) Three-story building structure, (b) Aluminum beams.

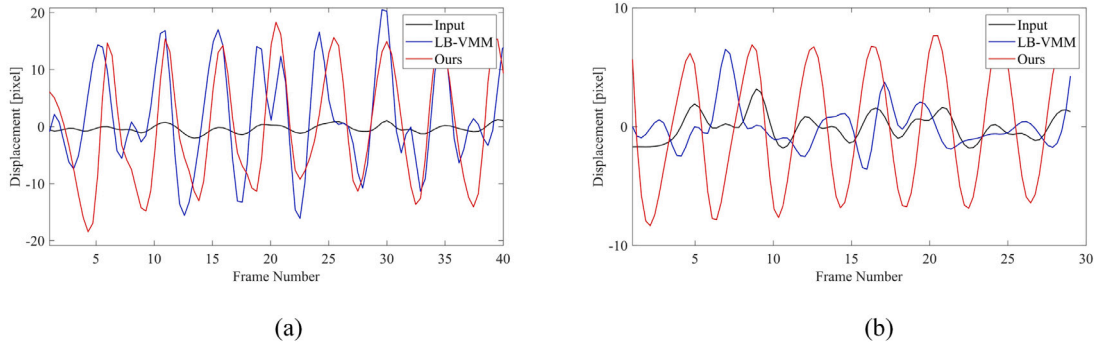
### 3.2. Experiments on three-story building structure and cantilever beam model

In order to verify the effectiveness of the proposed method for micro-amplitude vibration magnification and measurement of real structural elements and extraction of modal vibration shapes, relevant experiments have been carried out using a bench model of a three-story building structure (discrete type) and a bench model of an aluminum cantilever beam (continuous type) structure proposed by Los Alamos National Laboratory [13]. The height of each layer of the three-story building structure is 177 mm, the width of the bottom plate of each layer is 305 mm, and the shooting distance is 750 mm. The height of the aluminum cantilever beam structure is 700 mm, the width is 40 mm, the thickness is 2 mm, and the shooting distance is 500 mm. The three-story built-up column structure consists of aluminum columns and a concentrated mass plate at each floor, whose base is fixed to a heavy solid aluminum plate on the ground. An impact hammer was used to horizontally excite the structure at the top floor, an impact hammer was used to horizontally excite the beam near the fixed end. The structure was filmed using a fixed video camera and an accelerometer was installed to measure the acceleration. The video is preprocessed with a resolution of  $384 \times 216$ , then the video is subjected to video magnification to analysis the vibration signal and extract the vibration modes.

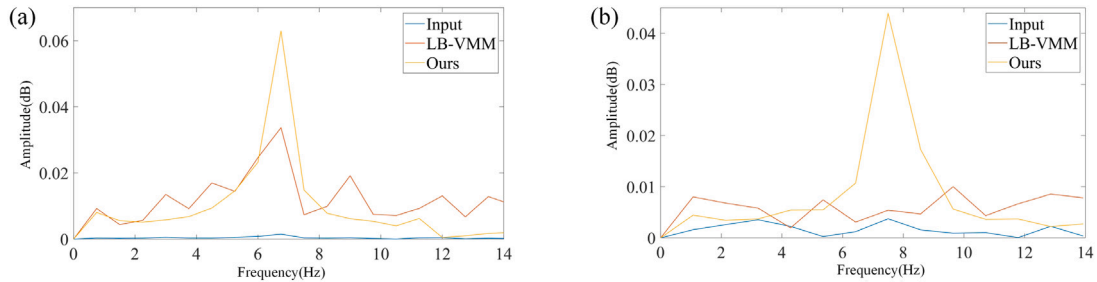
In this experiment we used the LB-VMM method [43,44] to compare and analysis with our proposed method. The LB-VMM model consists of three main components, encoder, manipulator, and decoder, which is based on the architecture proposed by Oh et al. [43,44] and has been implemented in the Pytorch machine learning framework. The synthetic dataset [43,44] was used to train the deep learning model. The Adam optimizer was used to optimize the network. During training, the learning rate was set to 0.0001, the default value of beta was (0.9, 0.999), and the Batch size was set to 6. Temporal filtering was used in the process of the phase-based method proposed in this paper, we set the band of interest to [5 Hz, 7 Hz] for the three-story building structure and [7 Hz, 8 Hz] for the aluminum cantilever beam structure, the optimized 1D Row GDGIF processing parameters are the same as above. We process the captured video sequences of micro-amplitude vibrations of the structure by training the LB-VMM model and our proposed video magnification methodology procedure to produce a series of magnified motion videos.

#### 3.2.1. Structural vibration measurement experiment

We conducted experiments aimed at extracting vibration information from video sequences. First, we conducted vibration displacement measurement experiments. We set the magnification factor to 50, as shown in Fig. 13, the displacement measurement area of the three-story building structure was set at the top floor, and the displacement measurement area of the aluminum cantilever beam structure was set at the top quarter part. We processed the original video using the method proposed in this paper and the LB-VMM method, respectively, and obtained the motion magnified video sequences. Subsequently, we used the displacement measurement method combining the Horn-Schunck method and the motion intensity averaging method proposed in this paper to process the motion-magnified video content. The results are shown in Fig. 14, which demonstrates the vibration displacement measurement results of a three-story building structure in Fig. 14(a) and the vibration displacement measurement results of an aluminum cantilever beam structure in Fig. 14(b).



**Fig. 14.** Comparison of vibration displacement results for three-story building structure and aluminum cantilever beam. (a) Three-story building structure, (b) Aluminum cantilever beam.



**Fig. 15.** Comparison of vibration frequency measurements of three-story building structures and aluminum cantilever beams. (a) Three-story building structure, (b) Aluminum cantilever beam.

The experimental results show that the two video magnification methods can magnify the micro-amplitude vibration of a three-story building structure with different qualities. However, the LB-VMM method exhibits irregular small wave peaks or troughs, which appear somewhat cluttered overall, indicating some high-frequency jitter and distortion in the structure. On the contrary, our method produced smooth displacement curves with more regular back and forth vibrations. Observing frame 20 in Fig. 14, we find a local minimum in the LB-VMM curve. In contrast, the proposed method for video magnification relies on phase information, and the magnified vibration displacements maintain a good correspondence with the extreme points of the input displacement curves. In the displacement measurements of the aluminum cantilever beam, the displacement curves generated by the LB-VMM were irregular and did not differ significantly from the amplitude of the vibration displacements in the original video, suggesting that the magnification of the micro-amplitude vibrations was not effective. The Proposed method provides regular and smooth vibration displacement curves, indicating more regular vibrations and smoother curves, verifying the effectiveness of our method in vibration displacement measurement of aluminum cantilever beams. In summary, the Proposed method shows better results in vibration displacement measurement of mechanical structures compared to previous methods, especially in magnifying the vibrations and keeping the vibration characteristics clear while reducing the structural deformation and information loss.

Based on the vibration frequency extraction method introduced in Section 2 of this paper, we carried out vibration frequency measurement experiments for a three-story building structure and an aluminum cantilever beam structure. The experimental results are illustrated in Fig. 15, where Fig. 15(a) shows the vibration frequency measurement results for the three-story building structure, while Fig. 15(b) shows the vibration frequency measurement results for the aluminum cantilever beam structure.

The experimental results show that both our proposed method and the LB-VMM method magnify the dominant frequency of the three-story building structure with different qualities and are consistent with the dominant frequency of the structure as estimated by the installed accelerometer [13]. However, the LB-VMM method magnifies the dominant frequency of the structure while the amplitudes of other frequencies show an increasing trend, compared to which our method obtains a higher amplitude of the dominant frequency and it is easier to differentiate it from the other frequencies. The LB-VMM method is not effective in terms of magnifying the dominant frequency of vibration of aluminum cantilever beams, whereas our method extracts the dominant frequency of vibration and improves the amplitude of the dominant frequency.

Vibration frequency extraction is essential for accurately assessing the health of a structure. Accurate measurements of the dominant frequency provide the fundamental vibration characteristics of the structure and are essential for detecting potential defects or abnormal behavior. Our method successfully magnifies the dominant frequencies of different structures, which is essential for accurately analyzing structural vibrations and capturing critical information for practical engineering applications. Therefore, these results provide reliable support for the application of our proposed method in the field of practical vibration measurement and structural health monitoring, especially in achieving the goal of primary frequency magnification while maintaining vibration accuracy.

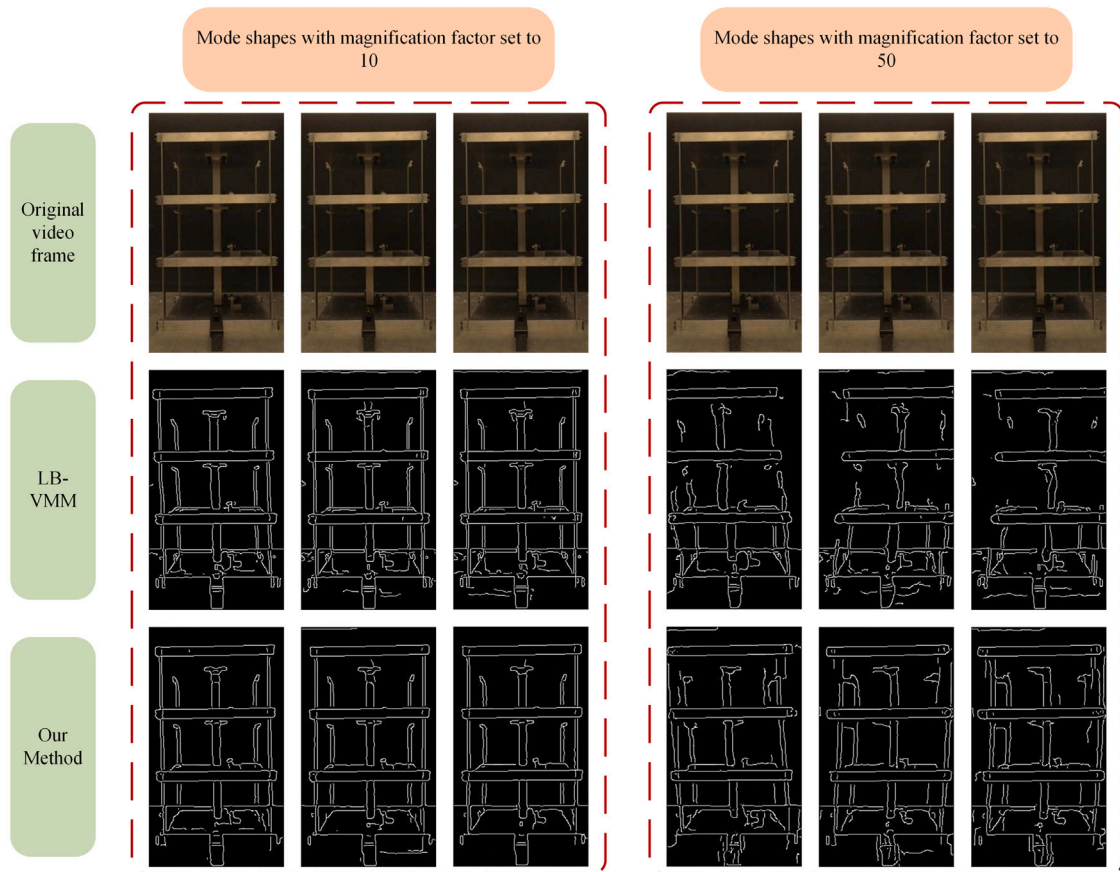


Fig. 16. Comparison of model shapes in the three-story building.

### 3.2.2. Model shapes extraction results

We proposed a model shapes measurement method that combines our proposed motion magnification approach with the Canny edge detection method. We compare this method with LB-VMM method to extract the modal shapes of three-story building structure and aluminum cantilever beam structure. We conducted two sets of experiments using magnification factors of 10 and 50, comparing the performance of these methods.

The experimental results, shown in Figs. 16 and 17, display a comparison of model shapes for the three-story building (Fig. 16) and the aluminum cantilever beam (Fig. 17). Each figure is divided into two parts representing magnification factors of 10 and 50. Within each part, there are three layers illustrating keyframe images from the original video, modal shape images obtained using the LB-VMM method, and modal shapes obtained using our proposed method. Compared to LB-VMM, our method exhibits a significant improvement in edge detection accuracy, particularly at higher magnification factors. These images illustrate the differences in modal shapes between the two methods at different magnification factors. LB-VMM introduces substantial distortion at higher magnification factors, impacting the extraction of modal shapes. This experiments highlight the advantages of our method, especially in minimizing information loss at higher magnification factors, resulting in clearer modal shapes with fewer distortions. This means that our method supports larger magnification factors and has the potential to enable magnification and non-contact measurement of more subtle vibrations.

## 4. Discussion and conclusion

In this work, we present a novel video motion magnification method designed to efficiently visualize and non-contact measurements of micro-amplitude vibrations. The method eliminates the need to physically connect expensive accelerometers and data acquisition equipment to a moving object, and it avoids the loss of structural information and artifacts in traditional video motion magnification methods.

To validate our approach, we built a vibration experimental platform and recorded videos capturing vibrations with different markers and frequencies. These videos underwent a series of pre-processing steps including phase extraction and magnification using a complex steerable pyramid, artifact and noise reduction via the optimized 1D Row GDGIF, and image pyramid reconstruction

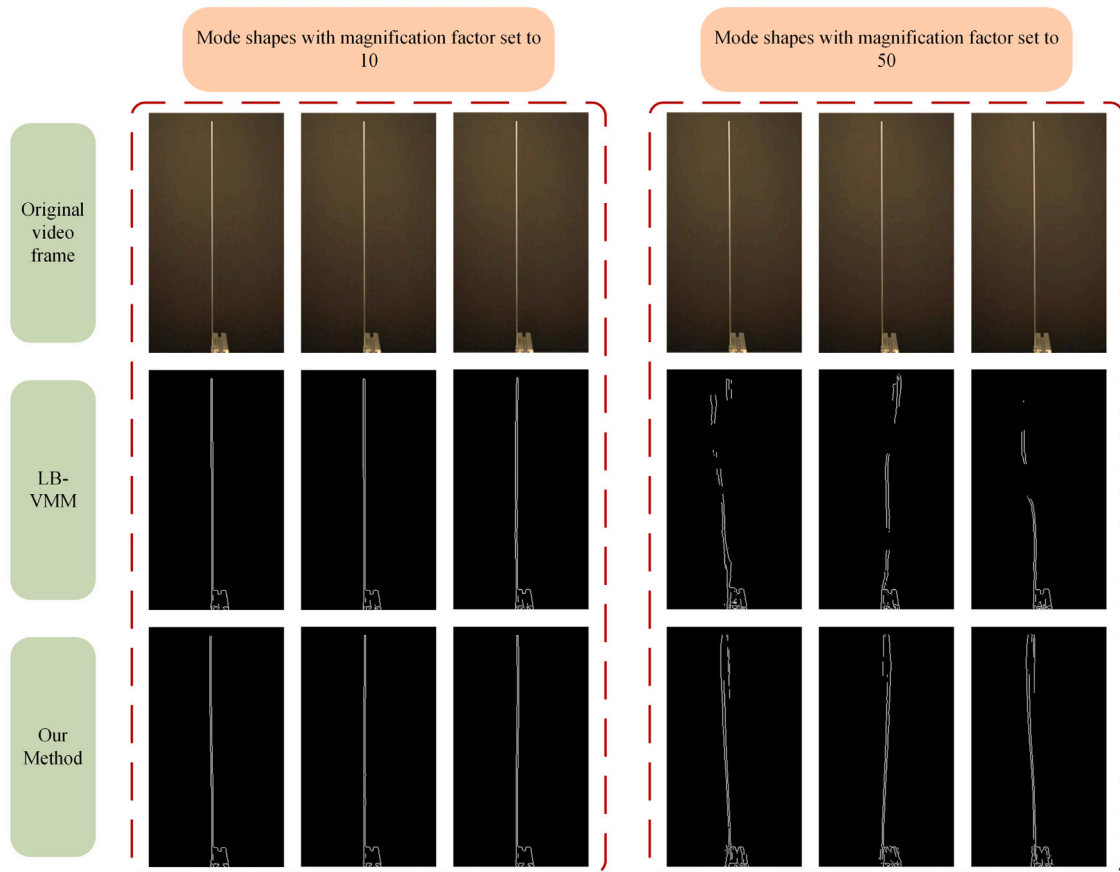


Fig. 17. Comparison of model shapes in the aluminum cantilever beam.

to produce motion magnification videos. We utilized spatio-temporal slicing to demonstrate the subjective improvement in motion magnification visibility achieved by our method. Additionally, we proposed a combination of the Horn–Schunck method and motion intensity averaging for displacement measurements and FFT frequency measurements for vibration analysis. We compare the video quality after motion magnification using objective quality evaluation metrics like SSIM and PSNR.

Experiments were conducted on both a three-story building structure and an aluminum cantilever beam to evaluate the efficacy of our proposed method in analyzing real structural elements vibrations. These experiments involved different aspects, including measuring vibration displacement, frequency, and extracting modal shapes. Our method significantly magnifies structural vibrations while accurately identifying dominant frequencies and maintaining their regularity and smoothness. Compared to existing video magnification techniques, our method demonstrates distinct advantages in structural vibration analysis, particularly in minimizing information loss and structural deformation while maintaining effective magnification of micro-amplitude vibrations. These findings underscore the promising practical applications of our method, especially in vibration measurement, structural health monitoring, and delivering precise and dependable analysis results.

This method effectively reduces artifacts generated during vibration motion magnification, significantly enhancing the precision of vibration displacement measurements. It selectively magnifies the dominant of primary vibration frequencies while preserving other amplitudes. Additionally, it demonstrates remarkable effectiveness in extracting modal vibration patterns. The proposed video micro-amplitude vibration magnification method offers advantages of global magnification, eliminating the need for tracking and region alignment. It efficiently magnifies micro-amplitude vibrations, providing higher-quality output videos for vibration analysis and enabling non-contact vibration measurements and modal vibration pattern acquisition.

The frame rate of about 30 frames per second for the experimental videos used in this paper can directly meet the needs of vibration frequency analysis of buildings, bridges, and similar structures [33] as higher frame rates increase computational complexity and resource requirements. Higher frequency microvibrations may exist in some practical situations. To overcome the limitations associated with the Nyquist theorem, we propose two potential solutions: algorithmic innovation in modal analysis and hardware enhancement. Innovations in modal analysis algorithms could enable vibration measurements at sampling frequencies lower than those traditionally recommended by the Shannon–Nyquist sampling theorem for output-only modal analysis. The spatio-temporal uncoupling property of the modal expansion allows for direct modal decoupling of temporally-aliased vibration

measurements, providing accurate mode shapes estimation [48]. Additionally, hardware integration, such as high-speed cameras have the potential to increase frame rates, enabling the capture of vibrating raw video at higher frequencies [31].

The proposed method employs a complex steerable pyramid, leading to a longer processing time compared to Eulerian linear magnification methods. In contrast to learning-based approaches, it demonstrates an advantage in terms of processing time as it does not require a training process and has lower hardware requirements. Processing the videos in this paper using the trained model requires a longer inference time than the proposed method in this paper. Real-time performance comparable to traditional PVMM methods.

Our experiments cover diverse camera-to-target distances, affirming the method's efficacy across various spatial scales. Its performance in situations where long distance non-contact vibration measurements are required can be further explored in the future. Focusing on applications in real structures with larger structural dimensions or larger distances will extend the application scenarios of the method.

Future work also could concentrate on algorithmic research to address video magnification's dependence on stable shooting environments. This would account for challenges such as camera shake and complex background motions, aiming to improve the accuracy of video magnification under unstable conditions.

### CRediT authorship contribution statement

**Yuanzhao Yang:** Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization.  
**Qi Jiang:** Project administration, Methodology, Funding acquisition.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

Data will be made available on request.

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